

**APPROXIMATION THEORY FOCUS GROUP - THE FIELDS INSTITUTE
2025 - PRESENTING H1 FOR COMPLETE FANS AND HYPERPLANE
ARRANGEMENTS, WORKING NOTES.**

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The following lemma is a useful presentation for $H_1(\mathcal{J}^{\mathbf{r}}[\Sigma])$ when Σ is complete. This is the analogue of [5, Lemma 3.8] for complete fans.

Lemma 0.1 ([1, Lemma 9.12]). *Let $\Sigma \subset \mathbb{R}^3$ be a hereditary, complete fan. Define $K^{\mathbf{r}} \subset \bigoplus_{\tau \in \Sigma_2} R(-\mathbf{r}(\tau) - 1)$ by*

$$K^{\mathbf{r}} = \left\{ \sum_{\tau \ni \gamma} f_{\tau} e_{\tau} \mid \gamma \in \Sigma_1, \sum_{\tau \ni \gamma} f_{\tau} \alpha_{\tau}^{\mathbf{r}(\tau)+1} = 0 \right\}.$$

Also define $V^{\mathbf{r}} \subset \bigoplus_{\tau \in \Sigma_2} R(-\mathbf{r}(\tau) - 1)$ by

$$V^{\mathbf{r}} = \left\{ \sum_{\tau \in \Sigma_2} f_{\tau} e_{\tau} \mid \sum_{\tau \in \Sigma_2} f_{\tau} \alpha_{\tau}^{\mathbf{r}(\tau)+1} = 0 \right\}.$$

Then $K^{\mathbf{r}} \subset V^{\mathbf{r}}$ and $H_1(\mathcal{J}^{\mathbf{r}}[\Sigma]) \cong V^{\mathbf{r}}/K^{\mathbf{r}}$ as R -modules.

Proof. The proof is similar to the proof of [5, Lemma 3.8]. Let $K_{\gamma}^{\mathbf{r}} \subset \bigoplus_{\gamma \in \tau} R(-\mathbf{r}(\tau))e_{\tau}$ be the module of relations around the ray $\gamma \in \Sigma_1$, namely

$$K_{\gamma}^{\mathbf{r}} = \left\{ \sum_{\tau \ni \gamma} f_{\tau} e_{\tau} \mid \sum_{\tau \ni \gamma} f_{\tau} \alpha_{\tau}^{\mathbf{r}(\tau)} = 0 \right\}.$$

Furthermore, let $J(\mathbf{0})$ be the ideal of the central vertex of Σ . Set up the following diagram with exact rows, whose first row is the complex $J[\Sigma]$.

$$\begin{array}{ccccc}
 & 0 & & 0 & & 0 \\
 & \uparrow & & \uparrow & & \uparrow \\
 \bigoplus_{\tau \in \Sigma_2} J^{\mathbf{r}}(\tau) & \longrightarrow & \bigoplus_{\gamma \in \Sigma_1} J^{\mathbf{r}}(\gamma) & \longrightarrow & J^{\mathbf{r}}(\mathbf{0}) \\
 & \uparrow & & \uparrow & & \uparrow \\
 \bigoplus_{\tau \in \Sigma_2} R(-\mathbf{r}(\tau)) & \longrightarrow & \bigoplus_{\substack{\gamma \in \Sigma_1, \tau \in \Sigma_2 \\ \gamma \in \tau}} R(-\mathbf{r}(\tau)) & \longrightarrow & \bigoplus_{\tau \in \Sigma_2} R(-\mathbf{r}(\tau)) \\
 & \uparrow & & \uparrow & & \uparrow \\
 0 & \longrightarrow & \bigoplus_{\gamma \in \Sigma_1} K_{\gamma}^{\mathbf{r}} & \xrightarrow{\iota} & V^{\mathbf{r}} \\
 & & \uparrow & & \uparrow \\
 & & 0 & & 0
 \end{array}$$

The middle row is in fact exact because the inclusion on the left hand side has the effect of gluing together copies of $R(-\mathbf{r}(\tau))$ that correspond to different rays in Σ_1 , leaving a copy of $R(-\mathbf{r}(\tau))$ for every codimension one face $\tau \in \Sigma_2$ in the cokernel. Now the long exact sequence in homology yields the isomorphisms $H_2(\mathcal{J}^\mathbf{r}[\Sigma]) \cong \ker(\iota)$ and $H_1(\mathcal{J}^\mathbf{r}[\Sigma]) \cong \operatorname{coker}(\iota)$. The image of $\bigoplus_{\gamma \in \Sigma_1} K_\gamma^\mathbf{r}$ under ι is precisely $K^\mathbf{r}$, so we are done. $\square$

Now suppose $\mathcal{A} = \bigcup_{i=1}^k H_i \subset \mathbb{R}^3$ is a hyperplane arrangement with associated complete fan $\Sigma^\mathcal{A}$. Let $\mathbf{r} : \Sigma_2^\mathcal{A} \rightarrow \mathbb{Z}_{\geq -1}$ be a smoothness distribution that is constant on hyperplanes (that is, if $\tau, \tau' \subset H \in \mathcal{A}$, then $\mathbf{r}(\tau) = \mathbf{r}(\tau')$). In this case, we also regard $\mathbf{r}$ as a map from $\mathcal{A} \rightarrow \mathbb{Z}_{\geq -1}$. Let $M^\mathbf{r} = [\alpha_1^{\mathbf{r}(H_1)} \dots \alpha_k^{\mathbf{r}(H_k)}]$ be the matrix whose entries are the linear forms defining the hyperplanes of $\mathcal{A}$, raised to the power stipulated by $\mathbf{r}$. Let

$$\operatorname{syz}(M^\mathbf{r}) := \left\{ \sum_{i=1}^k f_i e_i : \sum_{i=1}^k f_i \alpha_i^{\mathbf{r}(H_i)} = 0 \right\} \subset \bigoplus_{i=1}^k R(-\mathbf{r}(H_i))$$

be the syzygy module of the matrix $M^\mathbf{r}$.

For a given line W appearing as the intersection of at least two hyperplanes of $\mathcal{A}$, we write $M_W^\mathbf{r}$ for the matrix with a single row whose entries are $\{\alpha_H^{\mathbf{r}(H)} : W \subset H\}$. We similarly have

$$\operatorname{syz}(M_W^\mathbf{r}) := \left\{ \sum_{H \supset W} f_H e_H : \sum_{H \supset W} f_H \alpha_H^{\mathbf{r}(H)} = 0 \right\} \subset \bigoplus_{H \supset W} R(-\mathbf{r}(H) - 1).$$

There is a natural inclusion from $\operatorname{syz}(M_W^\mathbf{r})$ into $\operatorname{syz}(M^\mathbf{r})$ by extending the syzygy on $M_W^\mathbf{r}$ by zero to the rest of the entries of $\operatorname{syz}(M^\mathbf{r})$.

Corollary 0.2. *If $\mathcal{A} = \bigcup_{i=1}^k H_i \subset \mathbb{R}^3$ is a hyperplane arrangement with associated complete fan $\Sigma^\mathcal{A}$, and $\mathbf{r} : \Sigma_2^\mathcal{A} \rightarrow \mathbb{Z}_{\geq 0}$ is a smoothness distribution, then*

$$H_1(\mathcal{J}[\Sigma^\mathcal{A}]) \cong \frac{\operatorname{syz}(M^\mathbf{r})}{\sum_{W \in L_2(\mathcal{A})} \operatorname{syz}(M_W^\mathbf{r})},$$

where $L_2(\mathcal{A})$ is the collection of lines appearing as intersections of hyperplanes of $\mathcal{A}$.

1. THREE-GENERATED HYPERPLANE ARRANGEMENTS

The case $\mathbf{r} \equiv \mathbf{0}$ of Corollary 0.2 deserves special attention.

Corollary 1.1. *If $\mathcal{A} = \bigcup_{i=1}^k H_i \subset \mathbb{R}^3$ is a central and essential hyperplane arrangement with associated complete fan $\Sigma^\mathcal{A}$ and $\mathbf{r} \equiv \mathbf{0}$, then $H_1(\mathcal{J}[\Sigma^\mathcal{A}])$ is isomorphic to the $\mathbb{R}$ -vector space of $\mathbb{R}$ -linear relations among the linear forms $\alpha_1, \dots, \alpha_k$ modulo the $\mathbb{R}$ -linear relations among $\alpha_1, \dots, \alpha_k$ of length three.*

Proof. In this case, $\mathbf{r}(\tau) = 1$ for all $\tau \in \Sigma_2^\mathcal{A}$, so $M^\mathbf{r} = [\alpha_1 \dots \alpha_k]$. Suppose $\alpha_1, \alpha_2, \alpha_3$ are a basis for the $\mathbb{R}$ -span of the entries of $M^\mathbf{r}$ (this is three dimensional since $\mathcal{A}$ is essential).

Then $\operatorname{syz}(M^\mathbf{r})$ is generated by the Koszul syzygies on $\{\alpha_1, \alpha_2, \alpha_3\}$ along with all the $\mathbb{R}$ -linear relations on the entries of $M^\mathbf{r}$.

If $\bar{\gamma} \in L_2(\mathcal{A})$, then we can select two linear forms, without loss suppose these are α_1 and α_2 , that intersect in the line $\bar{\gamma}$. The syzygy module $\operatorname{syz}(M_{\bar{\gamma}}^\mathbf{r})$ is generated by the Koszul syzygy between α_1 and α_2 , along with the $\mathbb{R}$ -linear relations on $\{\alpha_H\}_{\gamma \in H}$. Since these linear forms effectively live in the two-dimensional vector space spanned by α_1 and α_2 , the relations among them all have length three.

From the above descriptions, we see that the Koszul syzygies in $\text{syz}(M^{\mathbf{r}})$ appear also in $\sum_{\bar{\gamma} \in L_2(\mathcal{A})} \text{syz}(M_{\bar{\gamma}}^{\mathbf{r}})$.

Thus the presentation in Corollary 0.2 implies that

$$H_1(\mathcal{J}[\Sigma^{\mathcal{A}}]) \cong \frac{\text{syzy}_0(M^{\mathbf{r}})}{\sum_{\bar{\gamma} \in L_2(\mathcal{A})} \text{syzy}_0(M_{\bar{\gamma}}^{\mathbf{r}})},$$

where syzy_0 represents ‘syzygies of degree zero’ – that is, $\mathbb{R}$ -linear relations.

Furthermore, any relation of length three among $\{\alpha_1, \dots, \alpha_k\}$, without loss suppose $c_1\alpha_1 + c_2\alpha_2 + c_3\alpha_3 = 0$, necessarily expresses the fact that α_1, α_2 , and α_3 all vanish along a common line $\bar{\gamma} \in L_2(\mathcal{A})$. Thus this relation appears in $\text{syzy}_0(M_{\bar{\gamma}}^{\mathbf{r}})$. It follows that we may recast the above presentation as the space of all $\mathbb{R}$ -linear relations on $\alpha_1, \dots, \alpha_k$ modulo the space of $\mathbb{R}$ -linear relations of length three. $\square$

Definition 1.2. If $\mathcal{A} = \bigcup_{i=1}^k H_i$ is a hyperplane arrangement with H_i the vanishing locus of α_i for $i = 1, \dots, k$, then $\mathcal{A}$ is called 3-generated if the space of all $\mathbb{R}$ -linear relations among $\alpha_1, \dots, \alpha_k$ is generated by the relations of length 3.

Lemma 1.3. If $\mathcal{A} = \bigcup_{i=1}^k H_i \subset \mathbb{R}^3$ is a hyperplane arrangement with associated complete fan $\Sigma^{\mathcal{A}}$, and $\mathbf{r} : \Sigma_2^{\mathcal{A}} \rightarrow \mathbb{Z}_{\geq 0}$ is a smoothness distribution, then $H_1(\mathcal{J}[\Sigma^{\mathcal{A}}])$ has finite length. Furthermore $S^{\mathbf{r}}(\Sigma^{\mathcal{A}})$ is free if and only if $H_1(\mathcal{J}[\Sigma^{\mathcal{A}}]) = 0$.

Sketch of proof. Show that the localization of the presentation in Corollary 0.2 at all homogeneous prime ideals besides the maximal ideal vanishes. The latter fact (concerning freeness) follows from a seminal result of Schenck [3], generalized in [4, Theorem 3.4]. In the three-dimensional case, this can be argued fairly quickly using Ext. $\square$

Corollary 1.4. $S^0(\Sigma^{\mathcal{A}})$ is free if and only if $\mathcal{A}$ is 3-generated.

The subtlety of this can be seen in action with an example that is sometimes called *Ziegler’s pair*. There will be a Macaulay demo walking through this example.

2. UNWINDING THE DIMENSION OF SPLINES ON COMPLETE THREE-DIMENSIONAL FANS

We regard $\mathbf{r}$ and Σ as understood in this section, and so we write $\mathcal{J}$ instead of $\mathcal{J}^{\mathbf{r}}[\Sigma]$. If Σ is a complete three-dimensional fan, Lemma 0.1 can be used to give an interesting dimension formula for $S^{\mathbf{r}}(\Sigma)$. Using the Euler characteristic of $\mathcal{J}$ and the fact that $H_0(\mathcal{J}) = 0$ we get

$$\dim H_2(\mathcal{J})_d - \dim H_1(\mathcal{J})_d = \sum_{\tau \in \Sigma_2} \dim J(\tau)_d - \sum_{\gamma \in \Sigma_1} J(\gamma)_d + \dim J(\mathbf{0})_d$$

Now recall that $S^{\mathbf{r}}(\Sigma) \cong R \oplus H_2(\mathcal{J})$. This gives

$$\dim S^{\mathbf{r}}(\Sigma)_d = \dim R_d + \sum_{\tau \in \Sigma_2} \dim J(\tau)_d - \sum_{\gamma \in \Sigma_1} \dim J(\gamma)_d + \dim J(\mathbf{0})_d + \dim H_1(\mathcal{J})_d. \quad (2.1)$$

Since $R = \mathbb{R}[x, y, z]$, $\dim R_d = \binom{d+2}{2}$. Since $J(\tau) = \langle \alpha_{\tau}^{\mathbf{r}(\tau)+1} \rangle$, $\dim J(\tau)_d = \binom{d+2-(\mathbf{r}(\tau)+1)}{2} = \binom{d+1-\mathbf{r}(\tau)}{2}$. The dimension of $J(\gamma)_d$ is computed by Schenck and Geramita in [2]. Their result is the following. $J(\gamma) = \langle \alpha_{\tau}^{\mathbf{r}(\tau)+1} : \tau \ni \gamma \rangle$ is a codimension two Cohen-Macaulay ideal, and thus it has a Hilbert-Burch resolution. Suppose that there are $n_{\gamma} + 1$ codimension one cones containing γ and that $\tau_0, \dots, \tau_{n_{\gamma}} \in \Sigma_2$ are these codimension one cones. Then $J(\gamma)$ is generated in degrees $e_0 = \mathbf{r}(\tau_0) + 1, \dots, e_{n_{\gamma}} = \mathbf{r}(\tau_{n_{\gamma}}) + 1$. Put $E_{\gamma} = \sum_{i=0}^{n_{\gamma}} e_i$. The key technical result of [2], which is Theorem 2.7 of that paper, is that $\text{syzy} J(\gamma)$ is generated in degrees $d_1, \dots, d_{n_{\gamma}}$ where $d_1 \leq d_2 \leq \dots \leq d_{n_{\gamma}}$ are each as close as possible to E_{γ}/n_{γ} . To state this exactly, let $E_{\gamma} = q_{\gamma}n_{\gamma} + r_{\gamma}$

be the result of dividing E_γ by n_γ using the Euclidean algorithm. Then $d_1 = d_2 = \dots = d_{n_\gamma - r_\gamma} = q_\gamma$ and $d_{n_\gamma - r_\gamma + 1} = \dots = d_{n_\gamma} = q_\gamma + 1$. Thus

$$\dim J(\gamma)_d = \sum_{\tau \ni \gamma} \binom{d+1 - \mathbf{r}(\tau)}{2} - (n_\gamma - r_\gamma) \binom{d+2 - q_\gamma}{2} - r_\gamma \binom{d+1 - q_\gamma}{2}.$$

In Equation (2.1), this leaves the term $\dim J(0)_d + \dim H_1(\mathcal{J})_d$ unsimplified. We simplify it as follows. Consider the (generally non-minimal) exact sequence

$$0 \rightarrow \text{syz}[\alpha_\tau^{\mathbf{r}(\tau)+1} : \tau \in \Sigma_2] \rightarrow \bigoplus_{\tau \in \Sigma_2} R(-\mathbf{r}(\tau) - 1) \xrightarrow{[\alpha_\tau^{\mathbf{r}(\tau)+1} : \tau \in \Sigma_2]} J(0) \rightarrow 0.$$

In the notation of Lemma 0.1, $\text{syz}[\alpha_\tau^{\mathbf{r}(\tau)+1} : \tau \in \Sigma_2] = V^{\mathbf{r}}$. Thus we have $\dim J(0) = \sum_{\tau \in \Sigma_2} \binom{d+1 - \mathbf{r}(\tau)}{2} - \dim V_d^{\mathbf{r}}$. By Lemma 0.1, we have $\dim H_1(\mathcal{J})_d = \dim V_d^{\mathbf{r}} - \dim K_d^{\mathbf{r}}$. Thus

$$\begin{aligned} \dim J(0)_d + \dim H_1(\mathcal{J})_d &= \left(\sum_{\tau \in \Sigma_2} \binom{d+1 - \mathbf{r}(\tau)}{2} - \dim V_d^{\mathbf{r}} \right) + (\dim V_d^{\mathbf{r}} - \dim K_d^{\mathbf{r}}) \\ &= \sum_{\tau \in \Sigma_2} \binom{d+1 - \mathbf{r}(\tau)}{2} - \dim K_d^{\mathbf{r}}. \end{aligned}$$

Using these expressions to simplify (2.1) and using the fact that $\sum_{\tau \in \Sigma_2} \sum_{\tau \ni \gamma} \binom{d+1 - \mathbf{r}(\tau)}{2} = 2 \sum_{\tau \in \Sigma_2} \binom{d+1 - \mathbf{r}(\tau)}{2}$ (each codimension one cone τ contains two rays) we get the following.

Theorem 2.1.

$$\dim S^{\mathbf{r}}(\Sigma)_d = \binom{d+2}{2} + \sum_{\gamma \in \Sigma_1} \left((n_\gamma - r_\gamma) \binom{d+2 - q_\gamma}{2} - r_\gamma \binom{d+1 - q_\gamma}{2} \right) - \dim K_d^{\mathbf{r}}.$$

Using Corollary 0.2, there is a different formulation of this theorem for a fan $\Sigma^{\mathcal{A}}$ induced by a hyperplane arrangement.

Theorem 2.2. *Let $\mathcal{A}$ be a central hyperplane arrangement and $\Sigma = \Sigma^{\mathcal{A}}$ its induced fan. Suppose that $\mathbf{r} : \Sigma_2 \rightarrow \mathbb{Z}_{\geq -1}$ is constant along hyperplanes of $\mathcal{A}$ (so we may regard $\mathbf{r}$ also as a function $\mathbf{r} : \mathcal{A} \rightarrow \mathbb{Z}_{\geq -1}$). Then*

$$\begin{aligned} \dim S^{\mathbf{r}}(\Sigma) &= \binom{d+2}{2} - \sum_{\tau \in \Sigma_2} \binom{d+1 - \mathbf{r}(\tau)}{2} + \sum_{\gamma \in \Sigma_1} \left((n_\gamma - r_\gamma) \binom{d+2 - q_\gamma}{2} - r_\gamma \binom{d+1 - q_\gamma}{2} \right) \\ &\quad + \sum_{H \in \mathcal{A}} \binom{d+1 - \mathbf{r}(H)}{2} - \dim \left(\sum_{W \in L_2(\mathcal{A})} \text{syz}(M_W^{\mathbf{r}}) \right) \end{aligned}$$

3. GENERIC HYPERPLANE ARRANGEMENTS

A hyperplane arrangement $\mathcal{A} = \cup_{i=1}^k H_i \subset \mathbb{R}^3$ is called *generic* if no three distinct hyperplanes $H_i, H_j, H_k \in \mathcal{A}$ intersect along a common line. In this case we have the following explicit formulas.

If $\gamma \in \Sigma_1^{\mathcal{A}}$, then there are exactly four two-dimensional cones of $\Sigma^{\mathcal{A}}$ that contain γ , and these four span only two distinct planes $H, H' \in \mathcal{A}$ by the assumption that $\mathcal{A}$ is generic. So $J(\gamma) = \langle \alpha_H^{\mathbf{r}(H)+1}, \alpha_{H'}^{\mathbf{r}(H')+1} \rangle$ is a complete intersection.

Moreover $\text{syz} M_W^{\mathbf{r}}$ is quite simple: each $W \in L_2(\mathcal{A})$ is the intersection of exactly two hyperplanes $H, H' \in \mathcal{A}$. Thus $\text{syz} M_W^{\mathbf{r}}$ consists of the Koszul syzygy between $\alpha_H^{\mathbf{r}(H)+1}$ and $\alpha_{H'}^{\mathbf{r}(H')+1}$. It follows that $\sum_{W \in L_2} \text{syz} M_W^{\mathbf{r}}$ is the image of the Koszul syzygies inside the free module $\bigoplus_{H \in \mathcal{A}} R(-\mathbf{r}(H) - 1)$.

Let us write $K_2(\mathcal{A}, \mathbf{r})$ for the $|\mathcal{A}| \times \binom{|\mathcal{A}|}{2}$ matrix whose columns are the Koszul syzygies between all pairs $\alpha_H^{\mathbf{r}(H)+1}, \alpha_{H'}^{\mathbf{r}(H')+1}$. That is, let $N = \bigoplus_{H \in \mathcal{A}} R(-\mathbf{r}(H) - 1)$. Then $K_2(\mathcal{A}, \mathbf{r})$ represents the map

$$\wedge^2 N \xrightarrow{K_2(\mathcal{A}, \mathbf{r})} N.$$

We have the following proposition.

Proposition 3.1. *If $\mathcal{A} \subset \mathbb{R}^3$ is a generic hyperplane arrangement and $\Sigma = \Sigma^{\mathcal{A}}$ the induced fan, then*

$$\begin{aligned} \dim S^{\mathbf{r}}(\Sigma^{\mathcal{A}})_d &= \binom{d+2}{2} + \sum_{\tau \in \Sigma_2} \binom{d+1-\mathbf{r}(\tau)}{2} \\ &\quad - \sum_{\substack{\gamma \in \Sigma_1 \\ \gamma \subset H \cap H'}} \left[\binom{d+1-\mathbf{r}(H)}{2} + \binom{d+1-\mathbf{r}(H')}{2} - \binom{d-\mathbf{r}(H)-\mathbf{r}(H')}{2} \right] \\ &\quad + \sum_{H \in \mathcal{A}} \binom{d+1-\mathbf{r}(H)}{2} - \dim(\text{im } K_2(\mathcal{A}, \mathbf{r}))_d \end{aligned}$$

This reduces finding $\dim S^{\mathbf{r}}(\Sigma^{\mathcal{A}})$ to finding the Hilbert function of the image of the Koszul matrix $K_2(\mathcal{A}, \mathbf{r})$.

Example 3.2. Suppose $\mathcal{A} = \cup_{i=1}^4 H_i$ is a generic three-dimensional arrangement and $\mathbf{r}(H_i) = 0$ for $1 \leq i \leq 4$. Let $\alpha_i = \alpha_{H_i}$ for $1 \leq i \leq 4$. Then

$$K_2(\mathcal{A}, \mathbf{r}) = \begin{array}{c} \begin{matrix} & 12 & 13 & 14 & 23 & 24 & 34 \end{matrix} \\ \begin{matrix} H_1 \\ H_2 \\ H_3 \\ H_4 \end{matrix} \begin{bmatrix} \alpha_2 & \alpha_3 & \alpha_4 & 0 & 0 & 0 \\ -\alpha_1 & 0 & 0 & \alpha_3 & \alpha_4 & 0 \\ 0 & -\alpha_1 & 0 & -\alpha_2 & 0 & \alpha_4 \\ 0 & 0 & -\alpha_1 & 0 & -\alpha_2 & -\alpha_3 \end{bmatrix} \end{array}.$$

This matrix is produced in Macaulay2 using the command
`koszul(matrix {{\alpha_1,\alpha_2,\alpha_3,\alpha_4}})`
 where $\alpha_1, \dots, \alpha_4$ are specific linear forms.

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